

Teaching Portfolio

Apurva Nakade, Assistant Teaching Professor, Department of Applied Mathematics & Statistics, Johns Hopkins University

Contents

Teaching philosophy	2
Courses	3
Discrete Mathematics JHU EN.553.171	3
Monte Carlo Methods JHU EN.553.433/633	4
Introduction to Computational Mathematics JHU EN.553.385	5
Intermediate Probability and Statistics JHU EN.553.311	6
Mathematical Foundations of AI JHU AS.110.110	6
Honors Algebra II, online JHU 412	7
Graph Theory JHU EN.553.672	7
Linear Algebra and Differential Equations JHU EN.553.291	8
Exploring Engineering Innovation JHU	8
MENU Linear Algebra and Multivariable Calculus Northwestern MATH 290	8
Single-Variable Differential Calculus Northwestern MATH 220-1	9
Introduction to Optimization Northwestern MATH 368	9
Foundations of Higher Mathematics Northwestern MATH 300	10
Elementary Differential Equations Northwestern MATH 250	10
Discrete Structures for Engineering Western Ontario MATH 2151A	10
Calculus I Western Ontario Calculus 1000A	11
Calculus II Western Ontario	11
Algebraic Topology Western Ontario	11
Topology Bootcamp Western Ontario	11
Topics in Category Theory Western Ontario	11
Honors Single Variable Calculus JHU AS.110.113	11
Differential Equations with Applications JHU AS.110.302	12
Symmetries and Polynomials JHU AS.110.361	12
Hitchhiker's Guide to Algebraic Topology JHU AS.110.360	12
Online Linear Algebra JHU AS.110.201	13
Mentoring	13
Professional development	13
Syllabi and sample materials	13
Reflection and goals	14

25	1,150+	3	2014	4.49 / 5
courses taught	students taught	universities	teaching since	mean JHU instructor rating, 2023-26

Teaching philosophy

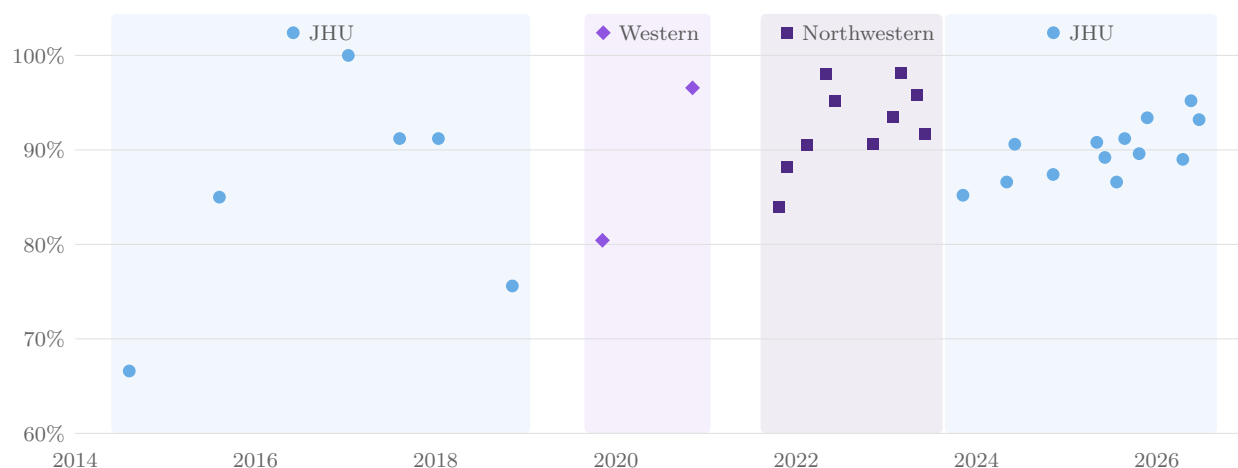
I have been teaching for more than a decade now. After all these years, I have come to understand that my role in the classroom and beyond is to support, motivate, and inspire students. I believe that the best way to learn mathematics is through making mistakes, getting confused, and struggling toward a solution. I consider myself a coach and a facilitator and teach with the core philosophy that my primary goal is to provide students with a welcoming and inclusive environment where experimentation is encouraged and honest mistakes aren't penalized. It is important to me that students who complete my courses leave with a sense of pride and accomplishment, and with increased interest in mathematics, curiosity, and self-confidence.

My teaching experience ranges from creating advanced electives and short bootcamp courses for small groups of students to managing and teaching in-person courses with several hundred students, coordinating multi-section classes, and adapting large service courses for asynchronous education. I have contributed to open source texts using technologies such as Webwork, Quarto, Pretext, and RMarkdown, and I'm involved in the long-term project of math formalization using the Lean theorem prover. I have taught topics spanning calculus, linear algebra, differential equations, discrete math, linear programming, math formalization, manifolds, algebraic topology, Monte Carlo methods, and computational math.

A central theme in my teaching has been adapting courses to student needs without watering down the content. I have redesigned most of the courses I teach, and a few ideas recur. I let students resubmit work to recover lost points, so they learn that mathematics is an iterative process rather than a rush to a final answer. I prefer frequent, smaller assessments to a few high-stakes exams. I write online-first course materials with code built in, and give older students final projects that offer them agency and a chance to showcase their skills. I also adapt courses to a post-AI world, deemphasizing what AI does better and encouraging students to use it to strengthen their learning. The course-by-course details are under Courses below.

- **Resubmissions:** students revise their work to recover lost points
- **Frequent, small assessments** instead of a few high-stakes exams
- **Online-first materials** with code built in
- **Final projects** that give students agency and a chance to showcase their skills
- **AI-aware design:** less of what AI does better, and AI used to strengthen learning

Courses



Instructor rating each semester, as a percentage of the top of each university's scale (5 at JHU, 6 at Northwestern, 7 at Western Ontario).

Discrete Mathematics JHU EN.553.171

Semester	Student level	Students	TAs	Course rating ¹	Instructor rating
Fall 2023	Undergraduate	56		4.11 / 5	4.26 / 5
Fall 2024	Undergraduate	41		4.05 / 5	4.37 / 5
Spring 2025	Undergraduate	25	4	4.32 / 5	4.54 / 5
Fall 2025	Undergraduate	45	4	4.33 / 5	4.48 / 5
Spring 2026	Undergraduate	30		4.31 / 5	4.45 / 5

Sample materials

- Course page, Fall 2026: syllabus, weekly schedule, lecture notes, and exam guides with topics, study advice and grading criteria

2025

- Rebuilt the course from scratch, drawing on teaching similar courses at Northwestern and Western Ontario during Covid: online video lectures, Webwork problems written from scratch, and an exam administered in Webwork
- Introduced homework resubmissions, so students can recover lost points and stop rushing to a final answer; mathematics is a creative, iterative process
- Replaced two long midterms with three in-class midterms, so students focus on a few topics at a time and a single bad score does not sink their grade
- Switched to an online textbook with built-in exercises; students now work regularly instead of only near deadlines and exams, and the grades of weaker students improved drastically

¹Each rating is the mean response to one of the two headline questions of the end-of-term student evaluations. At Johns Hopkins these are “The overall quality of this course is” and “The instructor’s teaching effectiveness is”, on a scale of 1 to 5; at Northwestern, “Provide an overall rating of the course” and “Provide an overall rating of the instruction”, on a scale of 1 to 6; at Western Ontario, “Course as a learning experience” and “Effective as a university teacher”, on a scale of 1 to 7. For a semester with several sections, the rating is the average over all respondents in all sections. The full reports are on GitHub.

- Introduced online assessments to improve feedback
- Several of the course’s TAs have gone on to TA other courses

Selected student comments

“The professor cares about the students and invests heavily in them. He is always very focused on the student’s learning evidenced by checking in with them during lecture to make sure they understand.”
(Spring 2026)

“The material is a range of topics that are fascinating and a solid overview of the AMS major. It was challenging but made me excited for future classes in AMS!”
(Fall 2025)

“Each lecture was interesting and presented in a very approachable way. The professor for this course was one of my favorites here at Hopkins thus far.”
(Fall 2025)

“One of the best aspects of this course is our professor’s exceptional teaching. The textbook can be limited and sometimes hard to understand, but our professor always explains difficult concepts in a way that is clear, logical, and easy to grasp. He chooses his words carefully and teaches with precision and structure.”
(Spring 2025)

“The grading system is very fair and gives you leniency on things like a missed homework. Dr Nakade actually cares about students’ understanding which sometimes feels rare at Hopkins.”
(Spring 2025)

“Dr. Nakade is an excellent teacher and really explained not only the material but how it was useful in different ways. He also seems to care greatly about the course and the students as well.”
(Fall 2024)

“Really interesting topics, fair and passionate instructor. This instructor did make me love math for the first time.”
(Fall 2023)

Monte Carlo Methods EN.553.433/633

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2024					
Spring 2025	Undergraduate and master’s	26	2	4.63 / 5	4.46 / 5
Fall 2025	Undergraduate	15	2	4.53 / 5	4.67 / 5
Spring 2026	Undergraduate and master’s	29		4.66 / 5	4.76 / 5

Sample materials

- Course page, Spring 2026: syllabus, course notes and course outline
- Final project: guidelines, rubrics, project ideas and past student project titles. Spring 2025 projects ranged from Hamiltonian Monte Carlo and adaptive rejection sampling to approximating PageRank and testing the randomness of LLM-generated numbers.

2025

- Updated the previous instructor’s course to align with the ways students learn today
- Rewrote course notes with an online-first approach, creating a course website instead of a traditional textbook

- Incorporated code directly into the online textbook for seamless integration of theory and practice
- Restructured weekly homework to include both programming and theory components: the theory explains the rationale behind the algorithms, and the programming immediately shows how to apply it
- Replaced traditional exams with weekly quizzes; the course covers fragmented topics with no central theory and nothing to memorize, and quizzes help students keep up with fast-paced material and reduce exam anxiety
- Added a final project, giving seniors and master's students agency in their learning and a way to showcase their skills; non-math majors have produced papers and software worthy of being presented at conferences
- Adapted the course for a post-AI world: students are encouraged to use AI assistants for programming assignments and final projects, but only their theoretical understanding is evaluated, and the final project requires a detailed report with the mathematical analysis
- Planning a CTEI grant application to update the online textbook and add interactive components

Selected student comments

“The course is taught very well and the course design is very good where you are learning something and then immediately being quizzed on it which really makes sure that you understand the content you are learning.” (Spring 2026)

“Wonderful course, all the content was very practical. In fact I used one of the topics in my paper right after being taught it (Bootstrapping).” (Fall 2025)

“Fantastic course, great teaching, super interesting content, great course structure. Love the quizzes, homework and presentation concept.” (Fall 2025)

“I really liked how the course was structured: weekly quizzes, interactive Jupyter notebooks, and a variety of various topics. The instructor did a really great job designing the weekly homeworks and quizzes, and it was very enjoyable to do the final project.” (Spring 2025)

“Fun topics, I liked that the course was centered more around learning rather than exams. I learnt a lot in a very unstressful environment.” (Spring 2025)

Introduction to Computational Mathematics EN.553.385

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2026	Undergraduate	49		4.48 / 5	4.66 / 5

Sample materials

- Course page, Spring 2026: syllabus, class notes and course outline
- Final project: guidelines, rubrics, project ideas and student project titles

Selected student comments

“Prof. Apurva is always open to reviewing even the most basic concepts and simplifies complicated concepts so it’s clear that he truly understands them. He does a great job engaging the class and encouraging participation. Even when I don’t know the answer, I feel safe enough to guess because I know he’ll find a way to validate the guess, so I don’t feel stupid in front of the class.” (Spring 2026)

“I really appreciated that there were no midterms and that homework was graded for completion. I felt like these policies reduced my stress a lot and made it much easier to learn a large amount of material.” (Spring 2026)

“You get to learn the part of math that they don’t really teach you: convergence and stability.” (Spring 2026)

Intermediate Probability and Statistics JHU EN.553.311

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2025	Undergraduate	19	0		
Summer 2025	Undergraduate	13	2	4.08 / 5	4.33 / 5

Selected student comments

“Professor Nakade is great, he cares about his students and it is much easier to learn from him.” (Summer 2025)

“The professor’s in-class problems and homework problems are extremely relevant to understanding the course, and are very helpful when studying for the exams.” (Summer 2025)

Mathematical Foundations of AI JHU AS.110.110

Semester	Student level	Students	TAs	Course rating	Instructor rating
Summer 2025	High school	36	1	4.44 / 5	4.56 / 5

Sample materials

- Course page, Summer 2026: daily outline, class notes and interactive tools used in class
- Course project: the three-part data analysis project. Summer 2025 groups predicted heart disease, bank customers’ decisions, exam performance and mushroom toxicity from public datasets; from Summer 2026, groups find their own real-world data.

Summer 2025

- Revamped the entire course to fit the audience of smart, motivated high school students with limited mathematical and programming background
- Made sessions interactive and flipped most classes
- Designed collaborative learning activities where students work with each other
- Incorporated discussions on topics like AI ethics, which got students hooked on the subject
- Developed a streamlined Python curriculum to teach programming from scratch and quickly transition to applications

Selected student comments

“I had never done coding before, and this course, although challenging at times, did teach me a lot and provided me with an excellent base.” (Summer 2025)

“How class time is split into two, with one part being lecture overall and the other being everyone coding individually. Also the professor. He goes around the classroom in the second half and helped everybody a lot with their codings and projects.” (Summer 2025)

“This course is very intellectually challenging, and I really enjoy the group discussions of the material, as they are very engaging and lead to more learning overall.” (Summer 2025)

“This course is definitely more fast pace and will often go past a lot of material very quickly. Despite that it was extremely enjoyable and satisfying and unlike any other course I’ve ever taken.” (Summer 2025)

Honors Algebra II, online [JHU](#) 412

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2024	Undergraduate	1	0		
Fall 2024	Undergraduate				
Spring 2025	Undergraduate				
Summer 2025	Undergraduate				

Graph Theory [JHU](#) EN.553.672

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2024	Graduate	25		4.32 / 5	4.33 / 5

Sample materials

- Course page, Spring 2024: weekly schedule with lecture notes and Jupyter notebooks
- Final projects: guidelines, topic suggestions and the projects presented. Topics ranged from Wagner’s theorem and perfect graphs to graph neural networks, phylogenetics and brain diffusion networks.

Selected student comments

“The instructor explains all the topics perfectly, and his passion for the same is clearly evident. The structure of the course is in such a way that you feel like you are solving puzzles all the time.” (Spring 2024)

“Instructor is enthusiastic and lectures are organized, material is explained very clearly. Nice breadth and depth of topics covered for a first course in graph theory. ... Final project in lieu of final exam was a great added learning experience.” (Spring 2024)

“Despite being a proof-based math class, there is not much assumed background to take this class. So, if you enjoy discrete math and brain teasers this is a perfect matching (no pun intended.)” (Spring 2024)

Linear Algebra and Differential Equations **JHU** EN.553.291

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2024	Undergraduate	56		4.17 / 5	4.53 / 5

Selected student comments

“Professor Nakade explains topics very clearly and class is very easy to follow, despite the content being very difficult. The lecture notes that students are given are also extremely helpful. Overall, the course is very well organized and engaging. It’s my favorite course this semester.” (Spring 2024)

“This course is extremely difficult, but the lectures are taught very well, in a clear manner. At no point do I feel lost in what is being explained.” (Spring 2024)

“There aren’t so many students in my session, so I can have very direct communications with the professor, and he remembers each of our names.” (Spring 2024)

Exploring Engineering Innovation **JHU**

Semester	Student level	Students	TAs	Course rating	Instructor rating
Summer 2024	High school				

MENU Linear Algebra and Multivariable Calculus **Northwestern** MATH 290

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2021 (290-1)	Undergraduate	19		4.88 / 6	5.29 / 6
Winter 2022 (290-2)	Undergraduate	18		5.21 / 6	5.43 / 6
Spring 2022 (290-3)	Undergraduate	16		5.21 / 6	5.71 / 6
Fall 2022 (290-1)	Undergraduate	46		5.17 / 6	5.44 / 6
Winter 2023 (290-2)	Undergraduate	24		5.44 / 6	5.89 / 6
Spring 2023 (290-3)	Undergraduate	17		5.14 / 6	5.50 / 6

2022-23

- Course coordinator

Selected student comments

“Apurva is probably the best teacher I’ve ever had. His class was my first at Northwestern, and ever since then I’ve enjoyed building a connection to him and just simply talking with him during office hours or outside of class. He is extremely intelligent and supportive, and I’ve grown a lot as a student from simply solving problems with him.” (Spring 2023)

“Apurva is doing great to constantly ask us for feedback and do what he can to improve the class however that may be. Just a great instructor! The best I’ve had yet!” (Winter 2023)

“Apurva is great at explaining concepts and he is very knowledgeable about every subject in the class as well. Additionally, he’s great at teaching and explaining mistakes to you while not making you feel bad about it.” (Fall 2022)

“I bombed the first midterm for not studying, and Apurva talked me through the possibilities and ultimately made me stay in the course.” (Fall 2022)

“Apurva is awesome!! Really funny and easily approachable/very available guy. He always stayed after class to answer questions for anyone who needed it.” (Winter 2022)

“Apurva is very encouraging when you are struggling with a problem and you can really tell that he was excited about math and teaching!” (Fall 2021)

Single-Variable Differential Calculus Northwestern MATH 220-1

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2021	Undergraduate	38		4.22 / 6	5.04 / 6

Selected student comments

“Apurva was a great professor. He explained everything very clearly and gave great lecture notes. In class we did lots of practice problems which was super helpful.” (Fall 2021)

“He was very patient in answering questions and was very good at helping you work through the problem to find exactly where you were getting stuck and then working through the rest. This way he was able to teach us what to do without doing it for us.” (Fall 2021)

Introduction to Optimization Northwestern MATH 368

Semester	Student level	Students	TAs	Course rating	Instructor rating
Winter 2022					
Spring 2022	Undergraduate and graduate	15		5.33 / 6	5.88 / 6

Selected student comments

“Apurva is phenomenal! He broke down key concepts with ease, and homework questions went over a variety of different examples. Optimization is interesting as a whole due to its wide applicability in other fields, but I felt this was an enjoyable course because of Apurva.” (Spring 2022)

“He explains difficult concepts in a digestible way, and makes sure everyone has the time and space to ask questions and apply the material they are learning.” (Spring 2022)

Foundations of Higher Mathematics **Northwestern** MATH 300

Semester	Student level	Students	TAs	Course rating	Instructor rating
Winter 2023	Undergraduate	24		5.28 / 6	5.61 / 6

Selected student comments

“Apurva is a great professor, and he really helped make the class enjoyable. He learned all of our names, and he was super approachable. His weekly office hours were super helpful for working through the homework and discussion worksheets. His lectures were really engaging, as he asked a lot of questions for the students to answer and work through.”
(Winter 2023)

Elementary Differential Equations **Northwestern** MATH 250

Semester	Student level	Students	TAs	Course rating	Instructor rating
Spring 2023	Undergraduate	17		5.42 / 6	5.75 / 6

Selected student comments

“Apurva was always willing to help, and also explained things very clearly. High quality prepared notes that were available to review later. Great energy for an early morning class. Obviously cared about the subject and student success.”
(Spring 2023)

Discrete Structures for Engineering **Western Ontario** MATH 2151A

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2020	Undergraduate	197		6.59 / 7	6.76 / 7

Selected student comments

“You are enthusiastic and while the course material was not my favourite, your teaching made it my favourite class!”
(Fall 2020)

“This course presented some difficult concepts, but Apurva explained everything really well and was very organized in his teaching. He’s also super understanding and it’s evident that he cares a lot about his students, a characteristic that, sadly, is not present in many professors.”
(Fall 2020)

“There really is no improvement I can find. The organization is perfect, the length of the lectures is perfect, the content inside the lectures themselves are clear and simple, the follow-up questions are perfect, and even the first midterm was extremely fair and was exactly what the prof said it would be like.”
(Fall 2020, midterm survey)

Calculus I **Western Ontario** Calculus 1000A

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2019	Undergraduate	177		5.58 / 7	5.63 / 7

Selected student comments

“Apurva Nakade has so far been the best professor I have had. I believe that his methods for teaching and leading a class is highly effective in terms of presenting ideas and concepts in a way that can be learned efficiently.”
(Fall 2019)

Calculus II **Western Ontario**

Semester	Student level	Students	TAs	Course rating	Instructor rating
Winter 2020	Undergraduate				

Algebraic Topology **Western Ontario**

Semester	Student level	Students	TAs	Course rating	Instructor rating
Winter 2021	Graduate				

Topology Bootcamp **Western Ontario**

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2020					

Topics in Category Theory **Western Ontario**

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2019					

Honors Single Variable Calculus **JHU** AS.110.113

Semester	Student level	Students	TAs	Course rating	Instructor rating
Fall 2017	Undergraduate				
Fall 2018	Undergraduate	9		4.00 / 5	3.78 / 5

Selected student comments

“Apurva is great. He answers any questions you have and always gives you feedback. If there is something you’re not understanding, he takes the time to walk you through it. He is very understanding that life still happens outside of class and is just a great teacher in general.” (Fall 2018)

“The course was heavily tailored to the students and was a good introduction to proofs. ... I understand how to write mathematical proofs much better and I feel more confident in my understanding of calculus.” (Fall 2018)

Differential Equations with Applications JHU AS.110.302

Semester	Student level	Students	TAs	Course rating	Instructor rating
Summer 2015	Undergraduate	9		4.25 / 5	4.25 / 5
Summer 2017	Undergraduate	11		4.33 / 5	4.56 / 5

Selected student comments

“Apurva is a great teacher, he knows how to present content in a succinct and easily absorbable way.” (Summer 2017)

“He inspired us to ask the important questions about math, to explore the foundations of mathematics, and to pursue further academic engagements in math, physics and engineering. He is an enlightened and devoted teacher and friend.” (Summer 2015)

Symmetries and Polynomials JHU AS.110.361

Semester	Student level	Students	TAs	Course rating	Instructor rating
Interession 2018	Undergraduate	10		4.44 / 5	4.56 / 5

Selected student comments

“The no homework and freeform style of working through worksheets that break up the proofs was really cool.” (Interession 2018)

Hitchhiker’s Guide to Algebraic Topology JHU AS.110.360

Semester	Student level	Students	TAs	Course rating	Instructor rating
Interession 2017	Undergraduate	11		4.89 / 5	5.00 / 5

Selected student comments

“Apurv is an engaging and funny instructor who is passionate about the subject. He doesn’t hesitate to stop and ask if anyone has a question before proceeding on through relatively dense mathematical material.” (Interession 2017)

“It’s great! Take it, especially if you’re not a math major but are interested in mathematical concepts.”
(Intersession 2017)

Online Linear Algebra **JHU** AS.110.201

Semester	Student level	Students	TAs	Course rating	Instructor rating
Summer 2014	Undergraduate	34		3.41 / 5	3.33 / 5

Selected student comments

“I wasn’t expecting an online course to be run so well. In many ways, it is comparable to an actual class. The lectures for interactive and helpful. Homework submission was easy and painless. The grading was fair.”
(Summer 2014)

Mentoring

Placeholder.

Professional development

I have completed a Faculty Forward Fellowship and a certification course at the Teaching Academy at JHU, where I learned about several important pedagogical concepts such as inquiry-based learning, backward course design, and learning objectives, which I regularly incorporate into my own teaching. I regularly attend workshops at the Center for Teaching Excellence and Innovation at JHU, and I am a member of the MAA’s Project NExT’20 cohort. Besides new skills, these workshops let me take on the role of a student and stay grounded.

- Faculty Forward Fellowship, JHU, 2025
- MAA Section NExT Fellow, MD-DC-VA Section, 2023-25
- Introduction to Education Research Workshop, JHU, 2023
- MSRI Critical Issues in Mathematics Education Workshop, 2022
- MAA Modeling Inspiration for Differential Equations Workshop, 2022
- MAA Project NExT Fellow, Brown’20 cohort, 2020
- Mathematical Association of America Fellow, 2020-Present
- Workshops by the Center for Teaching and Learning, UWO, 2019-2020
- Online Mastery Grading Workshop, 2019
- Teaching Academy Certification, JHU, 2019
- Science of Learning Symposium, JHU, 2014-2018

Syllabi and sample materials

Placeholder: links to course websites, syllabi, and a sample homework, quiz and project rubric.

Reflection and goals

I want to keep updating existing courses and create new ones that are first and foremost useful to my students, with interactive and engaging materials. I'm planning on applying for a CTEI grant for help with this.

I also want to broaden my work from teaching to mentoring and outreach. Finally, I want to engage more with the broader teaching community. I'm looking forward to the reduced course load, which will allow me to do this.